

ITM SKILLS UNIVERSITY
SCHOOL OF FUTURE TECH

Case Study Report

On

Hatchetfield Axe-Throwing & Recreation Venues

Campus LAN Segmentation, Trunking & Spanning Tree

Case Study No. 122

Computer Networks

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Tools Used: Cisco Packet Tracer

Project Type: Enterprise Campus LAN Design and Implementation

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1. ABSTRACT

This project redesigns the switching layer of Hatchetfield Axe-Throwing & Recreation Venues, a three-location business (Downtown, Westside, and Harbor District) whose sites are interconnected in a triangle over leased metro fiber. The existing network is a single flat broadcast domain, which causes broadcast storms on league nights and allows customers' personal devices to reach the lane-safety sensor systems.

The proposed design segments the network into five functional VLANs (Lane-Safety-Systems, Staff-POS, Admin, IT-Mgmt, and Guest) plus an unused native VLAN (999). All three inter-switch connections are secured as 802.1Q trunks with VLAN pruning, DTP disabled, and the unused VLAN 999 as the native VLAN to mitigate VLAN hopping. Inter-VLAN routing is provided by router-on-a-stick, with extended ACLs that guarantee the Guest VLAN can never reach the Lane-Safety-Systems VLAN.

The triangle topology creates a Layer-2 loop, so Rapid PVST+ is configured with SW-HarborDistrict as the root bridge (priority 4096). The two parallel links between SW-HarborDistrict and SW-Downtown are bundled into a single LACP EtherChannel, leaving exactly one blocked port in the whole triangle. Port security (maximum 2 MACs, violation restrict) protects Westside's guest access ports, VTP transparent mode is recommended, and a link-failure simulation documents how STP re-converges. The design was modeled in Cisco Packet Tracer.

2. INTRODUCTION

Recreation venues combine two very different kinds of network traffic on the same cabling: business-critical systems (point-of-sale, booking, lane scoring, and safety sensors) and untrusted public traffic (customer phones and laptops on lobby Wi-Fi). When both share a flat network, a broadcast storm affects everything and any customer device can attempt to talk to any system, including equipment that controls physical safety on the throwing lanes.

Hatchetfield operates three locations:

- Downtown
- Westside
- Harbor District (HQ finance/admin)

The three locations are connected in a triangle: each site has one switch, and each switch connects to the other two. A triangle gives every site two paths to every other site, which provides redundancy, but it also forms a Layer-2 loop that must be controlled by Spanning Tree Protocol. This report documents the complete redesign: VLAN segmentation, secure trunking, inter-VLAN routing with access policy, loop prevention with a deliberately chosen root bridge, link aggregation, Layer-2 port security, a VTP recommendation, and a simulated link failure.

3. PROBLEM STATEMENT

Hatchetfield Axe-Throwing & Recreation Venues shares a single booking and lane-scoring system across its three locations. Currently every device sits on one flat network. The IT contractor reports complaints about broadcast storms during league nights and about customers' personal devices being able to reach the lane-safety sensor systems.

The redesign of the switching layer must:

- Segment traffic with VLANs
- Secure the trunk links between sites
- Guarantee a loop-free topology

- Add link redundancy where it is cheap to do so
- Ensure the Guest network can never reach the Lane-Safety-Systems network

4. PROJECT OBJECTIVES

4.1 VLAN Scheme

Finalize the VLAN scheme (15, 25, 35, 45, 925, 999) and assign it to the correct switch ports at each location.

4.2 Secure Trunking

Configure 802.1Q trunks on all three inter-switch connections, using an unused VLAN as the native VLAN, and explain the security reason.

4.3 Inter-VLAN Routing

Provide router-on-a-stick routing so VLANs can reach each other where policy allows, Guest never reaching Lane-Safety-Systems, and verify end-to-end connectivity.

4.4 Loop Prevention

Configure Spanning Tree with SW-HarborDistrict as root bridge; show priority values and a port-role/port-state table for the triangle.

4.5 Link Aggregation

Bundle the two parallel links between SW-HarborDistrict and SW-Downtown into one EtherChannel and explain STP's view of it.

4.6 Port Security

Configure port security on Westside's Guest ports (maximum 2 MACs, violation restrict) and justify restrict over shutdown.

4.7 VTP Recommendation

Recommend and defend a VTP mode (server, client, or transparent) for the three-location chain.

4.8 Failure Simulation

Simulate a link failure between SW-HarborDistrict and SW-Westside and document, step by step, how STP re-converges and which port changes state.

5. NETWORK REQUIREMENTS ANALYSIS & DESIGN ASSUMPTIONS

5.1 VLAN Requirements

The case study specifies the following VLANs:

Table 1: VLAN requirements from the case study

VLAN ID	Name	Where it lives	Typical devices
15	Lane-Safety-Systems	Downtown + Westside	Lane sensors, scoring terminals
25	Staff-POS	All locations	POS terminals, staff tablets
35	Admin	Harbor District	HQ finance/admin
45	IT-Mgmt	All locations	Switch/AP management interfaces
925	Guest	Westside	Public lobby Wi-Fi
999	Native (unused)	All trunk links	Deliberately unused - mitigates VLAN hopping

5.2 Design Assumptions

The case study leaves some details open. The following assumptions were made, and the rest of the report is consistent with them:

- Each location has one Cisco Catalyst 2960-24TT access switch (24 x FastEthernet, 2 x GigabitEthernet): SW-Downtown, SW-Westside, and SW-HarborDistrict.
- One router, HD-Router, sits at Harbor District and connects to SW-HarborDistrict. It provides router-on-a-stick inter-VLAN routing for all sites. Placing it next to the root bridge keeps the routed path short.
- Each VLAN is given a /24 subnet from a private 10.122.0.0/16 block (122 = case study number). The case study does not state host counts, so /24 is used for simplicity and growth.
- The Guest VLAN's access ports at Westside are wired lobby jacks and kiosks. (An access point uplink would need a different port-security profile, see Section 13.)
- Rapid PVST+ is used instead of classic 802.1D because of its faster convergence, which matters for the link-failure scenario.
- Test hosts use static IP addresses so the verification does not depend on a DHCP server.

5.3 Inter-VLAN Access Policy

The case study requires that every VLAN can reach the others where policy allows, with Guest never reaching Lane-Safety-Systems. The policy implemented in this design is:

Table 2: Inter-VLAN access policy (source row to destination column)

Source \ Destination	15 Lane-Safety	25 Staff-POS	35 Admin	45 IT-Mgmt	925 Guest
15 Lane-Safety	-	Permit	Deny	Permit	Deny
25 Staff-POS	Permit	-	Permit	Permit	Deny
35 Admin	Deny	Permit	-	Permit	Deny
45 IT-Mgmt	Permit	Permit	Permit	-	Deny
925 Guest	Deny	Deny	Deny	Deny	-

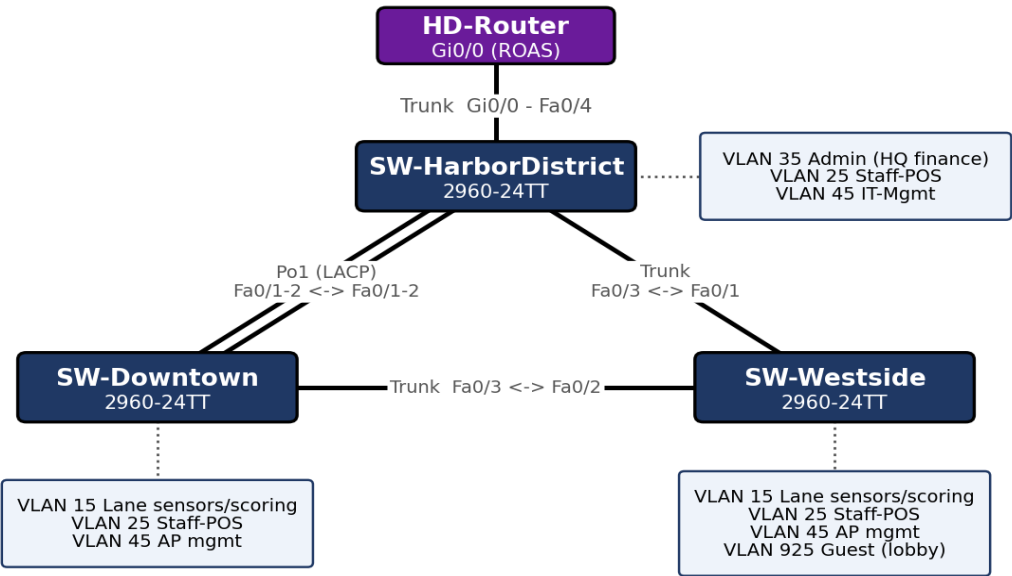
Rationale: Staff-POS and IT-Mgmt need to reach lane equipment (booking/scoring integration and administration). Admin (HQ finance) has no reason to reach safety sensors. Guests must reach nothing internal. IT-Mgmt may reach every internal VLAN for administration. Sessions from internal VLANs toward Guest are not provisioned, and Guest replies are dropped by the Guest ACL, so no such session can complete.

6. PHYSICAL NETWORK TOPOLOGY DESIGN

6.1 Topology Overview

The three switches form a triangle. SW-HarborDistrict and SW-Downtown are joined by two parallel links (later bundled into EtherChannel Po1), SW-HarborDistrict and SW-Westside by one link, and SW-Downtown and SW-Westside by one link. HD-Router connects to SW-HarborDistrict with a single trunk (the 'stick').

Hatchetfield Campus LAN - Physical Topology (Triangle)



All trunks: native VLAN 999 (unused), DTP disabled, VLAN 1 not allowed

Figure 1: Physical topology of the Hatchetfield campus LAN (triangle with EtherChannel and router-on-a-stick).

6.2 Link Inventory

Table 3: Physical links

Link	End A	End B	Type	Notes
L1	SW-HarborDistrict Fa0/1	SW-Downtown Fa0/1	Trunk	Member of Po1 (LACP)
L2	SW-HarborDistrict Fa0/2	SW-Downtown Fa0/2	Trunk	Member of Po1 (LACP)
L3	SW-HarborDistrict Fa0/3	SW-Westside Fa0/1	Trunk	Westside primary uplink
L4	SW-Downtown Fa0/3	SW-Westside Fa0/2	Trunk	Westside backup (STP-blocked)
L5	SW-HarborDistrict Fa0/4	HD-Router Gi0/0	Trunk	Router-on-a-stick

6.3 Packet Tracer Topology

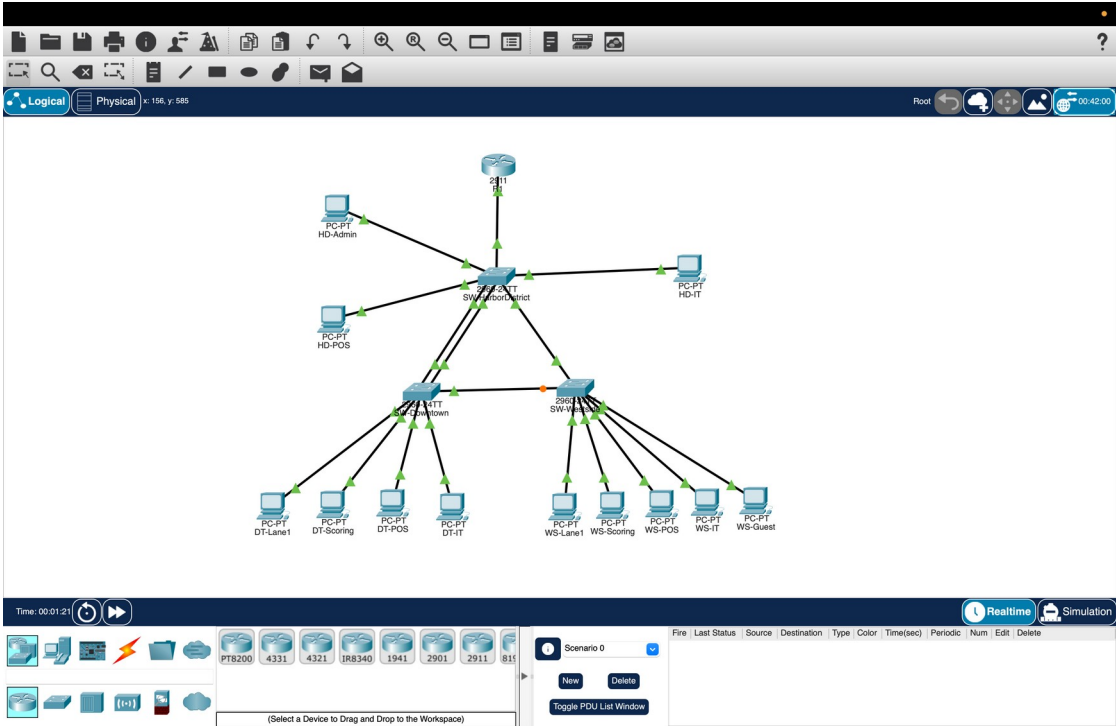


Figure 2: Packet Tracer logical topology of the Hatchetfield network.

6.4 Test Hosts

Table 4: End devices used for configuration and verification

Device	Switch / Port	VLAN	IP address	Gateway
LS-Sensor-DT-01	SW-Downtown Fa0/5	15	10.122.15.10/24	10.122.15.1
LS-Sensor-WS-01	SW-Westside Fa0/5	15	10.122.15.20/24	10.122.15.1
POS-DT-01	SW-Downtown Fa0/13	25	10.122.25.10/24	10.122.25.1
POS-WS-01	SW-Westside Fa0/11	25	10.122.25.20/24	10.122.25.1
POS-HD-01	SW-HarborDistrict Fa0/9	25	10.122.25.30/24	10.122.25.1
ADMIN-HD-01	SW-HarborDistrict Fa0/5	35	10.122.35.10/24	10.122.35.1
IT-PC-HD-01	SW-HarborDistrict Fa0/13	45	10.122.45.50/24	10.122.45.1
GUEST-WS-01	SW-Westside Fa0/17	925	10.122.200.10/24	10.122.200.1
GUEST-WS-02	SW-Westside Fa0/18	925	10.122.200.11/24	10.122.200.1

7. VLAN IMPLEMENTATION & PORT ASSIGNMENT

7.1 VLAN Database

VLAN names follow the case study. A VLAN must exist in a switch's database to be forwarded across its trunks, so VLANs 15 and 925 are also created on SW-HarborDistrict and 925 on SW-Downtown even though no access ports use them there: those switches carry the traffic in transit (for example, Westside traffic reaching the router at Harbor District). VLAN 35 (Admin) exists only at Harbor District because that is the only place it lives.

Table 5: VLANs created on each switch

VLAN	Name	SW-HarborDistrict	SW-Downtown	SW-Westside
15	Lane-Safety-Systems	Transit only	Access ports	Access ports
25	Staff-POS	Access ports	Access ports	Access ports
35	Admin	Access ports	Not created	Not created
45	IT-Mgmt	Access ports + SVI	Access ports + SVI	Access ports + SVI
925	Guest	Transit only	Transit only	Access ports
999	Native-Unused	Trunk native only	Trunk native only	Trunk native only

7.2 VLAN Assignment Table Mapped to Physical Ports

Table 6: VLAN-to-port assignment for all three switches

Switch	Ports	Mode	VLAN	Purpose
SW-HarborDistrict	Fa0/1-2	Trunk (Po1)	Native 999	EtherChannel to SW-Downtown
SW-HarborDistrict	Fa0/3	Trunk	Native 999	To SW-Westside Fa0/1
SW-HarborDistrict	Fa0/4	Trunk	Native 999	To HD-Router Gi0/0
SW-HarborDistrict	Fa0/5-8	Access	35 Admin	HQ finance/admin PCs
SW-HarborDistrict	Fa0/9-12	Access	25 Staff-POS	POS terminals, staff tablets
SW-HarborDistrict	Fa0/13-14	Access	45 IT-Mgmt	IT PCs / AP management
SW-HarborDistrict	Fa0/15-24, Gi0/1-2	Access (shutdown)	999	Unused
SW-Downtown	Fa0/1-2	Trunk (Po1)	Native 999	EtherChannel to SW-HarborDistrict
SW-Downtown	Fa0/3	Trunk	Native 999	To SW-Westside Fa0/2
SW-Downtown	Fa0/5-12	Access	15 Lane-Safety	Lane sensors, scoring terminals
SW-Downtown	Fa0/13-18	Access	25 Staff-POS	POS terminals, staff tablets
SW-Downtown	Fa0/19-20	Access	45 IT-Mgmt	AP management
SW-Downtown	Fa0/4, Fa0/21-24, Gi0/1-2	Access (shutdown)	999	Unused
SW-Westside	Fa0/1	Trunk	Native 999	To SW-HarborDistrict Fa0/3
SW-Westside	Fa0/2	Trunk	Native 999	To SW-Downtown Fa0/3

Switch	Ports	Mode	VLAN	Purpose
SW-Westside	Fa0/5-10	Access	15 Lane-Safety	Lane sensors, scoring terminals
SW-Westside	Fa0/11-14	Access	25 Staff-POS	POS terminals, staff tablets
SW-Westside	Fa0/15-16	Access	45 IT-Mgmt	AP management
SW-Westside	Fa0/17-24	Access + port-security	925 Guest	Public lobby ports
SW-Westside	Fa0/3-4, Gi0/1-2	Access (shutdown)	999	Unused

Security practice: Unused ports are placed in VLAN 999 and administratively shut down, so an unauthorized device plugged into a spare jack gets neither connectivity nor a path into any production VLAN.

7.3 Example: VLAN and Access-Port Configuration (SW-Westside)

```
SW-Westside(config)# vtp mode transparent
SW-Westside(config)# vlan 15
SW-Westside(config-vlan)# name Lane-Safety-Systems
SW-Westside(config-vlan)# vlan 25
SW-Westside(config-vlan)# name Staff-POS
SW-Westside(config-vlan)# vlan 45
SW-Westside(config-vlan)# name IT-Mgmt
SW-Westside(config-vlan)# vlan 925
SW-Westside(config-vlan)# name Guest
SW-Westside(config-vlan)# vlan 999
SW-Westside(config-vlan)# name Native-Unused
SW-Westside(config-vlan)# exit
SW-Westside(config)# interface range fa0/5 - 10
SW-Westside(config-if-range)# switchport mode access
SW-Westside(config-if-range)# switchport access vlan 15
SW-Westside(config-if-range)# spanning-tree portfast
```

Complete configurations for all three switches are in Appendix A.

(a) SW-HarborDistrict: show vlan brief

```
SW-HarborDistrict>enable
SW-HarborDistrict#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22
15	Lane-Safety-Systems	active	
25	Staff-POS	active	Fa0/2
35	Admin	active	Fa0/1
45	IT-Mgmt	active	Fa0/3
925	Guest	active	
999	Native	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

(b) SW-Downtown: show vlan brief

```
SW-Downtown>enable
SW-Downtown#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23
15	Lane-Safety-Systems	active	Fa0/1, Fa0/2
25	Staff-POS	active	Fa0/3
35	Admin	active	
45	IT-Mgmt	active	Fa0/4
925	Guest	active	
999	Native-Unused	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

(c) SW-Westside: show vlan brief

```
SW-Westside>enable
SW-Westside#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Gig0/1, Gig0/2
15	Lane-Safety-Systems	active	Fa0/1, Fa0/2
25	Staff-POS	active	Fa0/3
35	Admin	active	
45	IT-Mgmt	active	Fa0/4
925	Guest	active	Fa0/5
999	Native-Unused	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Figure 3: VLAN database and port membership on the three switches.

8. 802.1Q TRUNK CONFIGURATION & NATIVE VLAN

8.1 Trunk Design

All three inter-switch connections are 802.1Q trunks. (Link L5 to the router is also a trunk.) Each trunk is hardened with the same four settings:

- switchport mode trunk: the trunk is set statically, not negotiated.
- switchport nonegotiate: DTP is disabled so an attacker cannot negotiate a trunk on the port.
- switchport trunk native vlan 999: the native VLAN is the unused VLAN 999 on both ends.
- switchport trunk allowed vlan ...: only the VLANs that need to cross the link are allowed (VLAN pruning). VLAN 1 is not carried.

Table 7: Trunk allowed-VLAN lists

Trunk	Ends	Native	Allowed VLANs	Why
Po1 (L1+L2)	HD <-> DT	999	15,25,45,925,999	DT's VLANs plus Westside's VLANs in transit (failover). VLAN 35 not needed.
L3	HD <-> WS	999	15,25,45,925,999	All Westside VLANs. VLAN 35 not needed.
L4	DT <-> WS	999	15,25,45,925,999	Must carry Westside VLANs in case L3 fails, and Downtown VLANs if Po1 fails.
L5	HD <-> Router	999	15,25,35,45,925,999	The router needs a subinterface for every VLAN.

VLAN 35 (Admin) is pruned from every inter-switch trunk because it exists only at Harbor District. The other VLANs must be allowed on every triangle trunk, including the STP-blocked link L4, because after a link failure Spanning Tree will move traffic onto a path that was previously blocked. If that path did not carry the VLANs, failover would silently fail.

8.2 Trunk Configuration Example (SW-HarborDistrict)

```
SW-HarborDistrict(config)# interface fastEthernet0/3
SW-HarborDistrict(config-if)# description Trunk to SW-Westside Fa0/1
SW-HarborDistrict(config-if)# switchport mode trunk
SW-HarborDistrict(config-if)# switchport trunk native vlan 999
SW-HarborDistrict(config-if)# switchport trunk allowed vlan 15,25,45,925,999
SW-HarborDistrict(config-if)# switchport nonegotiate
```

(a) SW-HarborDistrict: show interfaces trunk

```
SW-HarborDistrict#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/23    on        802.1q         trunking    999
Fa0/24    on        802.1q         trunking    999

Port      Vlans allowed on trunk
Fa0/23    15,25,35,45,925,999
Fa0/24    15,25,35,45,925,999

Port      Vlans allowed and active in management domain
Fa0/23    15,25,35,45,925,999
Fa0/24    15,25,35,45,925,999

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/23    15,25,35,45,925,999
Fa0/24    15,25,35,45,925,999
```

(b) SW-Downtown: show interface trunk

```
SW-Downtown#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/24    on        802.1q         trunking    999

Port      Vlans allowed on trunk
Fa0/24    15,25,35,45,925,999

Port      Vlans allowed and active in management domain
Fa0/24    15,25,35,45,925,999

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/24    15,25,35,45,925,999
```

(c) SW-Westside: show interfaces trunk

```
SW-Westside#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/23    on        802.1q         trunking    999
Fa0/24    on        802.1q         trunking    999

Port      Vlans allowed on trunk
Fa0/23    15,25,35,45,925,999
Fa0/24    15,25,35,45,925,999

Port      Vlans allowed and active in management domain
Fa0/23    15,25,35,45,925,999
Fa0/24    15,25,35,45,925,999

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/23    15,25,35,45,925,999
Fa0/24    none
```

Figure 4: Trunk status, native VLAN and allowed VLANs.

8.3 Why the Native VLAN Is Set to an Unused VLAN (Security Justification)

On an 802.1Q trunk, frames that belong to the native VLAN are sent untagged, and by default the native VLAN is VLAN 1, which is also the default VLAN of every access port and the VLAN that carries control traffic such as CDP, DTP and VTP. This enables a VLAN-hopping attack called double tagging: an attacker whose access port is in the native VLAN sends a frame with two 802.1Q tags, an outer tag for the native VLAN and an inner tag for the victim VLAN (for example VLAN 15). The first switch sees the outer tag matches the native VLAN, strips it, and forwards the frame untagged onto the trunk; the next switch then reads the inner tag and delivers the frame into the victim VLAN, bypassing the router and every ACL. At Hatchetfield that could allow a customer

on a guest port to inject traffic into the lane-safety sensor VLAN. By setting the native VLAN to VLAN 999 on both ends of every trunk, and making sure VLAN 999 has no access ports, no SVI, no router IP address and no hosts, no attacker port can ever be a member of the native VLAN, so the double-tag trick has no starting point, and any untagged frame that does appear on a trunk lands in a VLAN that has nowhere to go. The native VLAN must be identical on both ends of each link (a mismatch triggers CDP warnings and STP inconsistencies), and it is combined here with `nonegotiate`, pruned allowed-VLAN lists without VLAN 1, and shutting down unused ports.

9. INTER-VLAN ROUTING (ROUTER-ON-A-STICK) & ACCESS POLICY

9.1 Router-on-a-Stick Design

HD-Router connects to SW-HarborDistrict Fa0/4 using a single physical interface (Gi0/0), which is divided into one 802.1Q subinterface per VLAN. Each subinterface is the default gateway of its VLAN. A subinterface for native VLAN 999 carries no IP address.

Table 8: HD-Router subinterfaces

Subinterface	VLAN	Encapsulation	IP address	Mask	ACL
Gi0/0.15	15	dot1Q 15	10.122.15.1	255.255.255.0	LANE_OUT (out)
Gi0/0.25	25	dot1Q 25	10.122.25.1	255.255.255.0	-
Gi0/0.35	35	dot1Q 35	10.122.35.1	255.255.255.0	-
Gi0/0.45	45	dot1Q 45	10.122.45.1	255.255.255.0	-
Gi0/0.925	925	dot1Q 925	10.122.200.1	255.255.255.0	GUEST_IN (in)
Gi0/0.999	999	dot1Q 999 native	none	-	-

```

HD-Router(config)# interface gigabitEthernet0/0
HD-Router(config-if)# no shutdown
HD-Router(config-if)# interface gigabitEthernet0/0.15
HD-Router(config-subif)# encapsulation dot1Q 15
HD-Router(config-subif)# ip address 10.122.15.1 255.255.255.0
HD-Router(config-subif)# interface gigabitEthernet0/0.999
HD-Router(config-subif)# encapsulation dot1Q 999 native
! (same pattern for VLANs 25, 35, 45, 925 - see Appendix A)

```

9.2 Access Control Lists

Two extended ACLs enforce the policy from Section 5.3. They give two independent layers of protection for the lane-safety VLAN:

- **GUEST_IN** (inbound on Gi0/0.925) drops everything the Guest VLAN sends toward any internal address (10.122.0.0/17) and permits the rest (for example, future Internet access).
- **LANE_OUT** (outbound on Gi0/0.15) allows only Staff-POS and IT-Mgmt to reach Lane-Safety-Systems and denies everything else. This protects the sensors from any source, not only Guest.

```

ip access-list extended GUEST_IN
deny ip 10.122.200.0 0.0.0.255 10.122.0.0 0.0.127.255
permit ip any any
ip access-list extended LANE_OUT
permit ip 10.122.25.0 0.0.0.255 10.122.15.0 0.0.0.255
permit ip 10.122.45.0 0.0.0.255 10.122.15.0 0.0.0.255
deny ip any any
!
interface gigabitEthernet0/0.925
ip access-group GUEST_IN in
interface gigabitEthernet0/0.15
ip access-group LANE_OUT out

```

Note: A Guest host can still ping its own gateway (10.122.200.1) because that address is outside 10.122.0.0/17. Guests are therefore connected to the network, yet isolated from every internal VLAN.

(a) HD-Router: show ip interface

```
R1>enable
R1#show ip interface
GigabitEthernet0/0 is up, line protocol is up (connected)
  Internet protocol processing disabled
GigabitEthernet0/0.15 is up, line protocol is up (connected)
  Internet address is 192.168.15.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound access list is not set
  Proxy ARP is enabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachable are always sent
  ICMP mask replies are never sent
  IP fast switching is disabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP Fast switching turbo vector
  IP multicast fast switching is disabled
  IP multicast distributed fast switching is disabled
  Router Discovery is disabled
  IP output packet accounting is disabled
  IP access violation accounting is disabled
  TCP/IP header compression is disabled
  RTP/IP header compression is disabled
  Probe proxy name replies are disabled
  Policy routing is disabled
  Network address translation is disabled
  BGP Policy Mapping is disabled
  Input features: MCI Check
  WCCP Redirect outbound is disabled
  WCCP Redirect inbound is disabled
  WCCP Redirect exclude is disabled
GigabitEthernet0/0.25 is up, line protocol is up (connected)
  Internet address is 192.168.25.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound access list is not set
  Proxy ARP is enabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachable are always sent
  ICMP mask replies are never sent
  IP fast switching is disabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP Fast switching turbo vector
  IP multicast fast switching is disabled
  IP multicast distributed fast switching is disabled
  Router Discovery is disabled
  IP output packet accounting is disabled
  IP access violation accounting is disabled
```

(b) HD-Router: show access-lists

```
R1#
R1#
R1#
R1#
R1#
R1#
R1#show access list
R1#show access-lists
Extended IP access list BLOCK_GUEST_TO_SAFETY
  10 deny ip 192.168.0.0 0.0.255.255 192.168.15.0 0.0.0.255
  20 permit ip any any
R1#
```

Figure 5: HD-Router subinterfaces and ACLs.

10. IP ADDRESSING DESIGN

All addresses are private (RFC 1918) and come from the 10.122.0.0/16 block. Internal VLANs use the VLAN ID as the third octet, which makes addresses easy to recognize during troubleshooting. The Guest VLAN is deliberately placed in a separate range (10.122.200.0/24) outside the internal block 10.122.0.0/17, so that one ACL line can match all internal space.

Table 9: VLAN subnets and gateways

VLAN	Name	Subnet	Mask	Gateway	Usable hosts
15	Lane-Safety-Systems	10.122.15.0/24	255.255.255.0	10.122.15.1	10.122.15.2 - .254
25	Staff-POS	10.122.25.0/24	255.255.255.0	10.122.25.1	10.122.25.2 - .254
35	Admin	10.122.35.0/24	255.255.255.0	10.122.35.1	10.122.35.2 - .254
45	IT-Mgmt	10.122.45.0/24	255.255.255.0	10.122.45.1	10.122.45.2 - .254
925	Guest	10.122.200.0/24	255.255.255.0	10.122.200.1	10.122.200.2 - .254

Table 10: Switch management addresses (SVI on VLAN 45)

Switch	Interface	IP address	Mask	Default gateway
SW-HarborDistrict	Vlan45	10.122.45.10	255.255.255.0	10.122.45.1
SW-Downtown	Vlan45	10.122.45.11	255.255.255.0	10.122.45.1
SW-Westside	Vlan45	10.122.45.12	255.255.255.0	10.122.45.1

Management SVIs are in IT-Mgmt (VLAN 45), the VLAN reserved for switch and AP management interfaces, so only the IT-Mgmt VLAN (and the router) can manage the switches. Each switch is reachable from every site through its trunks.

11. SPANNING TREE PROTOCOL DESIGN

11.1 The Layer-2 Loop

The three switches are connected in a triangle (and the HD-DT connection itself has two parallel links). Without Spanning Tree, a broadcast frame would circulate endlessly around the loop, flooding the network and corrupting MAC address tables, which would be even worse than the broadcast storms Hatchetfield already complains about. Spanning Tree Protocol removes the loop logically by blocking redundant ports while keeping them ready as backups.

11.2 Root Bridge Selection and Priority Values

The root bridge is the switch with the lowest Bridge ID (priority + VLAN ID as extended system ID + MAC address). Priorities must be multiples of 4096. SW-HarborDistrict is made the root bridge because it hosts the router (the busiest point of the network) and the HQ, so all traffic naturally flows toward the root. The other two switches are given explicit, distinct priorities so that results are predictable and do not depend on MAC address tie-breaks.

Table 11: STP root-bridge priority table

Switch	Configured priority	Effective Bridge ID priority (VLAN 25 example)	Role
SW-HarborDistrict	4096	$4096 + 25 = 4121$	Root bridge
SW-Downtown	8192	$8192 + 25 = 8217$	Secondary root (first backup)
SW-Westside	12288	$12288 + 25 = 12313$	Non-root

Configuration (applied to all VLANs that exist on each switch; the mode is Rapid PVST+):

```
! SW-HarborDistrict
spanning-tree mode rapid-pvst
spanning-tree vlan 15,25,35,45,925,999 priority 4096
! SW-Downtown
spanning-tree mode rapid-pvst
spanning-tree vlan 15,25,45,925,999 priority 8192
! SW-Westside
spanning-tree mode rapid-pvst
spanning-tree vlan 15,25,45,925,999 priority 12288
```

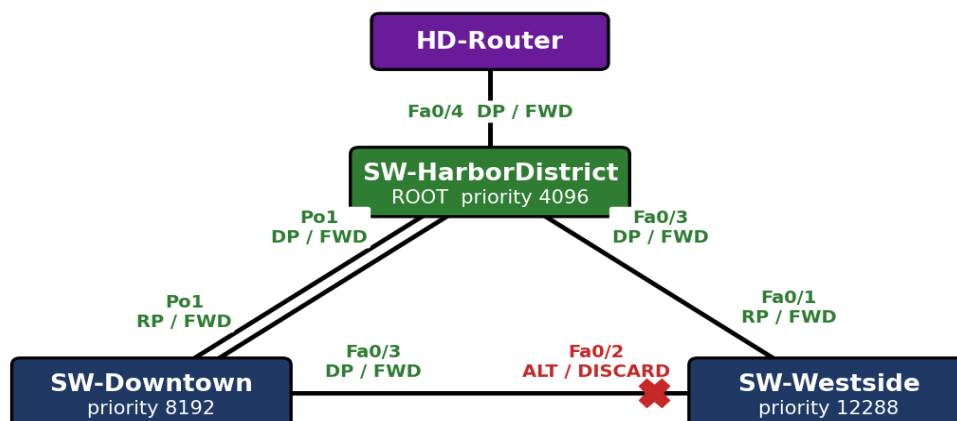
11.3 Port Roles and States in the Triangle

Port roles are decided in this order: (1) elect the root bridge; (2) each non-root switch picks the port with the lowest path cost to the root as its Root Port; (3) on every segment, the port with the lowest root path cost (tie-break: lowest bridge ID, then lowest port ID) becomes the Designated Port; (4) every remaining port becomes an Alternate port. Applying the rules to this topology:

1. SW-HarborDistrict is the root, so all of its ports are Designated Ports (Po1, Fa0/3 and Fa0/4).
2. SW-Downtown reaches the root directly through Po1. Because Po1 is an EtherChannel of two 100 Mbps links, its STP cost is lower than a single FastEthernet link, so Po1 is the Root Port.
3. SW-Westside reaches the root directly through Fa0/1 (cost 19). The other path, through SW-Downtown, costs more (Downtown's cost to root plus 19). Fa0/1 is the Root Port.
4. On the Downtown-Westside link, SW-Downtown advertises a root path cost lower than Westside's (its cost via Po1 is below 19, while Westside's is 19), so SW-Downtown's Fa0/3 is

the Designated Port. SW-Westside's Fa0/2 loses and becomes the Alternate port in the Discarding (blocking) state.

Rapid PVST+ Port Roles (EtherChannel active)



RP = Root Port DP = Designated Port ALT = Alternate (blocking) FWD = Forwarding

Only one port in the whole triangle is blocked: SW-Westside Fa0/2

Figure 6: Rapid PVST+ port roles and states in the triangle.

Table 12: Port role / port state table for every port in the triangle (EtherChannel active)

Switch	Port	Connected to	STP role	STP state
SW-HarborDistrict	Po1 (Fa0/1-2)	SW-Downtown Po1	Designated	Forwarding
SW-HarborDistrict	Fa0/3	SW-Westside Fa0/1	Designated	Forwarding
SW-HarborDistrict	Fa0/4	HD-Router Gi0/0	Designated	Forwarding
SW-Downtown	Po1 (Fa0/1-2)	SW-HarborDistrict Po1	Root	Forwarding
SW-Downtown	Fa0/3	SW-Westside Fa0/2	Designated	Forwarding
SW-Westside	Fa0/1	SW-HarborDistrict Fa0/3	Root	Forwarding
SW-Westside	Fa0/2	SW-Downtown Fa0/3	Alternate	Discarding (blocking)

Exactly one port in the whole triangle is blocked: SW-Westside Fa0/2. In Rapid PVST+ the 802.1D 'Blocking' state is called Discarding and the blocked role is called Alternate.

Before EtherChannel (for comparison)

If the two parallel HD-DT links are not bundled, STP sees them as two separate links and has to block one of them as well. With equal cost, SW-Downtown picks the port that receives the better BPDU from the root (the one connected to HarborDistrict's lowest port ID, Fa0/1) as its Root Port and blocks the other:

Table 13: Port roles before EtherChannel (two separate links)

Switch	Port	STP role	STP state
SW-HarborDistrict	Fa0/1, Fa0/2, Fa0/3	Designated	Forwarding
SW-Downtown	Fa0/1	Root	Forwarding
SW-Downtown	Fa0/2	Alternate	Discarding (blocking)
SW-Downtown	Fa0/3	Designated	Forwarding
SW-Westside	Fa0/1	Root	Forwarding
SW-Westside	Fa0/2	Alternate	Discarding (blocking)

In that case two ports (Downtown Fa0/2 and Westside Fa0/2) are blocked, and one of the two parallel cables carries no traffic. EtherChannel removes one of those blocked ports and puts the second cable to work.

(a) SW-HarborDistrict: show spanning-tree vlan 25 (root bridge)

```
SW-HarborDistrict#show spanning-tree vlan 25
VLAN0025
  Spanning tree enabled protocol ieee
  Root ID    Priority    4121
             Address     0030.F276.A174
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4121 (priority 4096 sys-id-ext 25)
             Address     0030.F276.A174
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	----	----	-----	-----	-----
Po1	Desg	FWD	3	128.27	P2p
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/23	Desg	FWD	19	128.23	P2p
Fa0/24	Desg	FWD	19	128.24	P2p

(b) SW-Westside: show spanning-tree vlan 25

```
SW-Westside#show spanning-tree vlan 25
VLAN0025
  Spanning tree enabled protocol ieee
  Root ID    Priority    4121
             Address     0030.F276.A174
             Cost         19
             Port         23 (FastEthernet0/23)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32793 (priority 32768 sys-id-ext 25)
             Address     000B.BE03.02E1
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	----	----	-----	-----	-----
Fa0/3	Desg	FWD	19	128.3	P2p
Fa0/23	Root	FWD	19	128.23	P2p
Fa0/24	Altn	BLK	19	128.24	P2p

(c) SW-HarborDistrict: show spanning-tree summary

```
SW-HarborDistrict#show spanning-tree summary
Switch is in pvst mode
Root bridge for: Lane-Safety-Systems Staff-POS Admin IT-Mgmt Guest Native
Extended system ID is enabled
Portfast Default is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default is disabled
EtherChannel misconfig guard is disabled
UplinkFast is disabled
BackboneFast is disabled
Configured Pathcost method used is short
```

Name	Blocking	Listening	Learning	Forwarding	STP Active
-----	-----	-----	-----	-----	-----
VLAN0015	0	0	0	5	5
VLAN0025	0	0	0	6	6
VLAN0035	0	0	0	6	6
VLAN0045	0	0	0	6	6
VLAN0925	0	0	0	5	5
VLAN0999	0	0	0	5	5
-----	-----	-----	-----	-----	-----
7 vlans	0	0	0	33	33

Figure 7: Spanning Tree output proving root bridge and port roles.

12. ETHERCHANNEL (HARBOR DISTRICT - DOWNTOWN)

12.1 Configuration

Fa0/1 and Fa0/2 on both SW-HarborDistrict and SW-Downtown are bundled into Port-channel 1 using LACP (IEEE 802.3ad) in active mode on both sides. The trunk settings are applied to the port-channel interface, and the member ports inherit them. Both members must have identical speed, duplex, and VLAN settings.

```
! Same commands on SW-HarborDistrict and SW-Downtown
interface range fastEthernet0/1 - 2
  channel-protocol lacp
  channel-group 1 mode active
exit
interface port-channel 1
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 15,25,45,925,999
  switchport nonegotiate
```

Packet Tracer tip: If your Packet Tracer version rejects 'channel-protocol lacp' or 'switchport nonegotiate', remove only that line. The design and results are unaffected. Configure both ends before expecting the bundle to come up, and use the same mode (active/active, or desirable/desirable with PAgP) on both ends.

12.2 What Happens to STP's View of the Connection

Once the EtherChannel is active, Spanning Tree no longer sees two parallel links between SW-HarborDistrict and SW-Downtown. It sees one logical port, Port-channel 1. The consequences are:

- STP runs on Po1, not on Fa0/1 and Fa0/2. BPDUs are sent and received over the logical port (physically over one member link), and the role and state shown by 'show spanning-tree' belong to Po1.
- No member link is blocked. Before bundling, STP had to block one of the two parallel links to break the loop between them. After bundling there is only one logical link between the two switches, so both cables forward traffic at the same time. Frames are load-balanced across members by a hash of address fields, and a given flow always uses the same member, so no loop and no frame duplication occur.
- The STP path cost is based on the aggregate bandwidth. Two 100 Mbps links give a lower cost than one, which is why Po1 is the preferred root path for Downtown.
- A single member failure is not a topology change. If one cable fails, the bundle keeps running on the remaining link at half the bandwidth; STP does not recalculate and no TCN is generated. Only if both cables fail does the connection go down and STP reconverge.

```
show etherchannel summary on SW-HarborDistrict
SW-HarborDistrict#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  S - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP       Gig0/1(P) Gig0/2(P)
```

Figure 8: EtherChannel Po1 bundle status.

13. PORT SECURITY ON THE GUEST VLAN

13.1 Configuration

Port security is applied to Westside's Guest VLAN access ports (Fa0/17-24): a maximum of 2 learned MAC addresses per port, and the violation action is restrict.

```
SW-Westside(config)# interface range fastEthernet0/17 - 24
SW-Westside(config-if-range)# switchport mode access
SW-Westside(config-if-range)# switchport access vlan 925
SW-Westside(config-if-range)# switchport port-security
SW-Westside(config-if-range)# switchport port-security maximum 2
SW-Westside(config-if-range)# switchport port-security violation restrict
SW-Westside(config-if-range)# switchport port-security mac-address sticky
SW-Westside(config-if-range)# spanning-tree portfast
SW-Westside(config-if-range)# spanning-tree bpduguard enable
```

Sticky learning records the first two MAC addresses seen on a port. Any frame from a third source MAC address is a violation. PortFast brings guest ports to forwarding immediately, and BPDU guard shuts a port down if it ever receives a BPDU, which stops a guest from plugging in a switch and interfering with the STP topology.

13.2 Justification: Restrict Instead of Shutdown for a Public Lobby

In 'shutdown' mode, a violation puts the port into the err-disabled state, which cuts the port completely and requires a person to clear it (shutdown / no shutdown or an errdisable recovery timer). In a public lobby with no technician on site, that behavior is a liability: a guest who legitimately connects a third device (for example a laptop, a phone and a tablet through a hub) would knock out the port for themselves and for the next customer, and anyone could cause a denial of service on purpose by simply plugging in extra devices. In 'restrict' mode the switch silently discards frames from the unauthorized MAC address, increments the violation counter and generates a syslog/SNMP message, while the port stays up and the two already-learned devices keep working. Restrict therefore still enforces the security goal (an attacker cannot use more than two MAC addresses on a port or flood the CAM table), keeps the lobby service available, and leaves an audit trail for the IT contractor. The trade-off is that the alert is easy to miss if nobody monitors the logs, so the violation counter should be reviewed ('show port-security interface') or forwarded to a monitoring system.

Limitation: If the lobby Wi-Fi access point's own uplink were plugged into one of these ports, every wireless client's MAC would appear on that single port and the limit of 2 would block them. The AP uplink should be on a separate port (for example in VLAN 45 or a trunk with its own profile).

show port-security address on SW-Westside

```
SW-Westside#show port-security address
      Secure Mac Address Table
-----
Vlan    Mac Address      Type                Ports    Remaining Age
-----
Total Addresses in System (excluding one mac per port)    : 0
Max Addresses limit in System (excluding one mac per port) : 1024
SW-Westside#
```

Figure 9: Port-security status on a Guest port.

14. VTP MODE RECOMMENDATION MEMO

TO:	Hatchetfield IT Contractor / Operations Management
FROM:	Yuvraj Mishra, Network Design
DATE:	October 2026
SUBJECT:	Recommended VTP mode for the three-location switch chain (Case Study 122)

Recommendation: run all three switches in VTP transparent mode (vtp mode transparent) and manage the VLAN database manually on each switch.

Why

VTP saves typing the same VLAN on many switches. Hatchetfield has three switches and about six VLANs that rarely change, so the saving is minimal while the failure mode is severe.

The risk we are avoiding

In server/client mode, every switch in the same VTP domain accepts the VLAN database from whichever switch has the highest revision number. If a spare or lab switch with a higher revision number and a different database is cabled in, it overwrites the VLAN database everywhere: VLANs 15, 25 and 925 could vanish in seconds, taking down lane safety, POS and guest access at all three sites at once, with nobody on site to notice. A VTP server also exposes the VLAN database to any device that can form a trunk, which conflicts with the hardened trunks in Section 8.

The trade-off we accept

With transparent mode, VLANs must be created by hand on all three switches, so a typo or a forgotten VLAN on one switch can cause a mismatch. That risk is local, happens only when someone makes a change, and is easy to audit by comparing 'show vlan brief' on the three switches. It is far smaller than an instant, network-wide overwrite.

Alternative that could also be defended

A VTP server at SW-HarborDistrict with clients at the other two sites is reasonable if Hatchetfield expects many more switches and frequent VLAN changes. It would need VTP version 3 (or at least a domain password) and a strict rule that any new switch is reset to revision 0 before cabling. For three switches, that discipline is not worth the risk.

Operational controls

- Add each new VLAN to all three switches (and to trunk allowed lists) in one change ticket, then verify with 'show vlan brief'.
- Keep the VLAN table in this report as the single source of truth.
- Verify with 'show vtp status' that VTP Operating Mode is Transparent on every switch.

15. LINK-FAILURE CONVERGENCE REPORT

15.1 Scenario

Link L3 between SW-HarborDistrict (Fa0/3) and SW-Westside (Fa0/1) fails. This is Westside's Root Port and its primary path to the root bridge and the router. The test uses a continuous ping from a Westside host to its gateway so that traffic loss during convergence can be observed.

15.2 Procedure

1. Confirm the starting state: 'show spanning-tree vlan 25' on SW-Westside shows Fa0/1 = Root/Forwarding and Fa0/2 = Alternate/Discarding.
2. On LS-Sensor-WS-01 open the command prompt and start a continuous ping: ping -t 10.122.15.1
3. On SW-HarborDistrict, shut down Fa0/3 (interface fastEthernet0/3, then shutdown). This simulates a cable or fiber failure. (In Packet Tracer you can also delete the cable.)
4. Observe the ping and run 'show spanning-tree vlan 25' on SW-Westside.
5. Restore the link (no shutdown) and observe the port roles return to normal.

15.3 Step-by-Step Re-convergence

Table 14: How Rapid PVST+ re-converges after the failure

Step	Event	What happens
0	Before failure	SW-Westside: Fa0/1 Root/Forwarding, Fa0/2 Alternate/Discarding. SW-Downtown Fa0/3 is Designated/Forwarding and keeps sending BPDUs to Westside, so Fa0/2 holds a stored BPDU describing the path to the root through Downtown.
1	Link goes down	SW-HarborDistrict Fa0/3 and SW-Westside Fa0/1 detect loss of link immediately (physical link down; no need to wait for the max-age timer).
2	Root Port lost	SW-Westside no longer has a Root Port. It runs STP and selects the best remaining path to the root, which is the stored BPDU on Fa0/2 (root path = Downtown's cost to root + 19).
3	Alternate becomes Root	Fa0/2 changes from Alternate/Discarding to Root/Forwarding. In RSTP an Alternate port can take over immediately as the new Root Port when the old Root Port has failed, because no loop is possible, so there are no listening/learning delays.
4	Topology change	SW-Westside sends BPDUs with the Topology Change flag; SW-Downtown and SW-HarborDistrict flush the MAC addresses learned on the affected ports. Frames to unknown destinations are briefly flooded until MAC tables are relearned.
5	New steady state	Traffic from Westside flows Fa0/2 -> SW-Downtown Fa0/3 -> Po1 -> SW-HarborDistrict -> HD-Router. SW-Downtown's Fa0/3 stays Designated/Forwarding. No other port changes state.

Table 15: Port states before and after the failure

Switch / Port	Before failure	After failure
SW-HarborDistrict Fa0/3	Designated / Forwarding	Down
SW-Westside Fa0/1	Root / Forwarding	Down
SW-Westside Fa0/2	Alternate / Discarding	Root / Forwarding (the only STP state change)
SW-Downtown Fa0/3	Designated / Forwarding	Designated / Forwarding (unchanged)
SW-Downtown Po1	Root / Forwarding	Root / Forwarding (unchanged)

15.4 Result and Comparison

The only port that changes STP state is SW-Westside Fa0/2 (Alternate/Discarding to Root/Forwarding). With Rapid PVST+ this is expected to complete in well under a few seconds, so the continuous ping should lose at most a few packets before replies resume. With classic 802.1D STP the same port would have to pass through Listening (15 s) and Learning (15 s) before forwarding, an outage of roughly 30 seconds, which is a good reason for choosing Rapid PVST+.

When the failed link is restored, HarborDistrict Fa0/3 again offers Westside a better (cheaper, direct) path. Fa0/1 becomes the Root Port again, Fa0/2 returns to Alternate/Discarding, and the topology returns to the state shown in Section 11.

16. NETWORK TESTING AND VERIFICATION

The design is verified at Layer 2 (VLANs, trunks, EtherChannel, STP, port security) and Layer 3 (inter-VLAN routing and ACL policy). The table lists each test, the expected result, and the figure where the Packet Tracer evidence should be placed.

16.1 Layer-2 Verification Commands

Table 16: Layer-2 verification checklist

Command	Device	What to confirm
show vlan brief	All switches	VLAN names and port membership match Section 7.2
show interfaces trunk	All switches	Native VLAN 999; allowed lists match Section 8.1; status trunking
show etherchannel summary	HD, DT	Po1 in use (SU), Fa0/1 and Fa0/2 flagged (P)
show spanning-tree vlan 25	All switches	HD is the root; WS Fa0/2 is Altn/blocked
show spanning-tree summary	HD	Mode rapid-pvst; root bridge for all VLANs
show port-security interface fa0/17	WS	Max 2, mode Restrict, violation count increases
show vtp status	All switches	Operating mode Transparent

16.2 Connectivity and Policy Tests

Table 17: End-to-end connectivity and policy test matrix

#	Source	Destination	Expected	Evidence
1	LS-Sensor-DT-01 (VLAN 15, Downtown)	10.122.15.1 (gateway)	Success	Fig. 11
2	LS-Sensor-DT-01	LS-Sensor-WS-01 (VLAN 15, Westside)	Success (same VLAN across switches)	Fig. 11
3	POS-DT-01	POS-WS-01	Success	Fig. 11
4	POS-DT-01 (Staff-POS)	LS-Sensor-WS-01 (Lane-Safety)	Success (policy permits)	Fig. 12
5	IT-PC-HD-01 (IT-Mgmt)	LS-Sensor-DT-01, POS-WS-01, ADMIN-HD-01	Success (IT reaches all)	Fig. 12
6	ADMIN-HD-01 (Admin)	POS-HD-01 (Staff-POS)	Success	Fig. 12
7	ADMIN-HD-01 (Admin)	LS-Sensor-DT-01 (Lane-Safety)	Fail (LANE_OUT)	Fig. 13
8	GUEST-WS-01 (Guest)	LS-Sensor-WS-01 (Lane-Safety)	Fail (GUEST_IN + LANE_OUT)	Fig. 13
9	GUEST-WS-01	POS, Admin, IT-Mgmt hosts	Fail (GUEST_IN)	Fig. 13
10	GUEST-WS-01	10.122.200.1 (own gateway)	Success	Fig. 13
11	Third device on Westside Fa0/17	-	Frames dropped, port stays up, counter increments	Fig. 9

For blocked tests, Packet Tracer shows 'Destination host unreachable' (the router answers with an ICMP administratively-prohibited message) or 'Request timed out'. Either one is a pass for a test that is expected to fail.

17. PROTOCOL AND OSI MODEL MAPPING

Table 18: Technologies used in this project mapped to the OSI model

Technology	OSI layer	TCP/IP layer	Role in this project
VLAN / 802.1Q tagging	Layer 2 - Data Link	Network Access	Segments broadcast domains and carries multiple VLANs over trunk links.
Spanning Tree (Rapid PVST+)	Layer 2 - Data Link	Network Access	Prevents switching loops in the triangle; provides redundancy.
EtherChannel / LACP	Layer 2 - Data Link	Network Access	Bundles two physical links into one logical link.
Port security	Layer 2 - Data Link	Network Access	Limits MAC addresses per guest port.
VTP / DTP	Layer 2 - Data Link	Network Access	VLAN database distribution and trunk negotiation (VTP transparent, DTP disabled).
Router-on-a-stick, IPv4	Layer 3 - Network	Internet	Routes between VLAN subnets.
Extended ACLs	Layers 3-4	Internet / Transport	Filter traffic by source and destination IP.
ICMP (ping)	Layer 3 - Network	Internet	Connectivity verification.

Table 19: Devices and their primary OSI operating layer

Device	Primary layer	Role
End devices (sensors, POS, PCs)	Layers 1-7	Generate and receive application traffic.
Catalyst 2960 switches	Layer 2	VLAN enforcement, MAC-based forwarding, STP, port security.
HD-Router	Layer 3	Inter-VLAN routing and ACL enforcement.

18. DELIVERABLES CHECKLIST

Table 20: Case study deliverables and where they are covered

Deliverable	Where
VLAN assignment table mapped to physical switch ports	Section 7.2
Trunk and EtherChannel configuration listings for all three switches	Sections 8, 12 and Appendix A
Inter-VLAN routing configuration (router-on-a-stick sub-interfaces)	Section 9 and Appendix A
STP root-bridge priority table and port-role/port-state table	Section 11.2 and 11.3
Port security configuration for the Guest VLAN with justification paragraph	Section 13
One-page VTP mode recommendation memo	Section 14
Short convergence report for the simulated link-failure scenario	Section 15
Native VLAN security paragraph (Objective 2)	Section 8.3
Explanation of STP's view of the EtherChannel (Objective 5)	Section 12.2

19. CONCLUSION

The Hatchetfield campus LAN was redesigned to replace a single flat network with a segmented, loop-free, redundant switching layer. Five functional VLANs separate lane-safety systems, staff and POS, administration, IT management and guest traffic, and the unused VLAN 999 is used as the native VLAN on every trunk to defend against VLAN hopping.

Router-on-a-stick with two extended ACLs enforces the policy that Guest can never reach Lane-Safety-Systems, and also restricts Lane-Safety-Systems to Staff-POS and IT-Mgmt only. Rapid PVST+ with SW-HarborDistrict as root bridge (priority 4096) removes the Layer-2 loop and leaves a single blocked port, SW-Westside Fa0/2, as a ready backup path. Bundling the two parallel HarborDistrict-Downtown links into an LACP EtherChannel doubles the usable bandwidth and eliminates a second blocked port. Port security in restrict mode protects the public lobby without making the ports easy to knock offline, and VTP transparent mode removes the risk of a network-wide VLAN database overwrite.

The link-failure simulation shows that only one port, SW-Westside Fa0/2, changes state when the HarborDistrict-Westside link fails, and that Rapid PVST+ restores connectivity within seconds rather than the roughly 30 seconds of classic STP.

20. FUTURE IMPROVEMENTS

- Replace router-on-a-stick with SVIs on a Layer-3 switch (or a Layer-3 core) so inter-VLAN routing no longer depends on a single 100 Mbps/1 Gbps trunk.
- Add a second router/Layer-3 device with first-hop redundancy (HSRP/VRRP) so there is no single point of failure for routing.
- Enable DHCP snooping, Dynamic ARP Inspection and IP Source Guard on the Guest VLAN access ports.
- Restrict management access: SSH only, an ACL on the VTY lines, and AAA (RADIUS/TACACS+) for switch administration.
- Restrict IT-Mgmt reachability and apply stateful filtering with a firewall at the edge.
- Use wireless controllers with WPA3-Enterprise for staff and a captive portal for guest Wi-Fi.
- Add monitoring (syslog, SNMP, NetFlow) so port-security violations and STP topology changes generate alerts.
- Upgrade the inter-site links to Gigabit EtherChannel and use BPDU guard/root guard on edge ports.

APPENDIX A: COMPLETE DEVICE CONFIGURATIONS

The listings below can be pasted into Cisco Packet Tracer's CLI tab. Paste the whole listing for a device in one go. Configure both ends of the EtherChannel before checking its status.

A.1 SW-HarborDistrict

```
enable
configure terminal
hostname SW-HarborDistrict
no ip domain-lookup
!
vtp mode transparent
!
vlan 15
  name Lane-Safety-Systems
vlan 25
  name Staff-POS
vlan 35
  name Admin
vlan 45
  name IT-Mgmt
vlan 925
  name Guest
vlan 999
  name Native-Unused
exit
!
spanning-tree mode rapid-pvst
spanning-tree vlan 15,25,35,45,925,999 priority 4096
!
interface range fastEthernet0/1 - 2
  description EtherChannel member to SW-Downtown Fa0/1-2
  channel-protocol lacp
  channel-group 1 mode active
  exit
interface port-channel 1
  description Trunk to SW-Downtown (EtherChannel Po1)
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 15,25,45,925,999
  switchport nonegotiate
  exit
interface fastEthernet0/3
  description Trunk to SW-Westside Fa0/1
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 15,25,45,925,999
  switchport nonegotiate
  exit
interface fastEthernet0/4
  description Trunk to HD-Router Gi0/0 (router-on-a-stick)
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 15,25,35,45,925,999
  switchport nonegotiate
  exit
!
interface range fastEthernet0/5 - 8
```

```
description Admin (HQ finance/admin)
switchport mode access
switchport access vlan 35
spanning-tree portfast
exit
interface range fastEthernet0/9 - 12
description Staff-POS
switchport mode access
switchport access vlan 25
spanning-tree portfast
exit
interface range fastEthernet0/13 - 14
description IT-Mgmt
switchport mode access
switchport access vlan 45
spanning-tree portfast
exit
interface range fastEthernet0/15 - 24 , gigabitEthernet0/1 - 2
description UNUSED
switchport mode access
switchport access vlan 999
shutdown
exit
!
interface vlan 45
ip address 10.122.45.10 255.255.255.0
no shutdown
exit
ip default-gateway 10.122.45.1
end
write memory
```

A.2 SW-Downtown

```
enable
configure terminal
hostname SW-Downtown
no ip domain-lookup
!
vtp mode transparent
!
vlan 15
name Lane-Safety-Systems
vlan 25
name Staff-POS
vlan 45
name IT-Mgmt
vlan 925
name Guest
vlan 999
name Native-Unused
exit
!
spanning-tree mode rapid-pvst
spanning-tree vlan 15,25,45,925,999 priority 8192
!
interface range fastEthernet0/1 - 2
description EtherChannel member to SW-HarborDistrict Fa0/1-2
channel-protocol lacp
```

```
channel-group 1 mode active
exit
interface port-channel 1
description Trunk to SW-HarborDistrict (EtherChannel Po1)
switchport mode trunk
switchport trunk native vlan 999
switchport trunk allowed vlan 15,25,45,925,999
switchport nonegotiate
exit
interface fastEthernet0/3
description Trunk to SW-Westside Fa0/2
switchport mode trunk
switchport trunk native vlan 999
switchport trunk allowed vlan 15,25,45,925,999
switchport nonegotiate
exit
!
interface range fastEthernet0/5 - 12
description Lane-Safety-Systems (sensors, scoring terminals)
switchport mode access
switchport access vlan 15
spanning-tree portfast
exit
interface range fastEthernet0/13 - 18
description Staff-POS
switchport mode access
switchport access vlan 25
spanning-tree portfast
exit
interface range fastEthernet0/19 - 20
description IT-Mgmt (AP management)
switchport mode access
switchport access vlan 45
exit
interface range fastEthernet0/4 , fastEthernet0/21 - 24 , gigabitEthernet0/1 - 2
description UNUSED
switchport mode access
switchport access vlan 999
shutdown
exit
!
interface vlan 45
ip address 10.122.45.11 255.255.255.0
no shutdown
exit
ip default-gateway 10.122.45.1
end
write memory
```

A.3 SW-Westside

```
enable
configure terminal
hostname SW-Westside
no ip domain-lookup
!
vtp mode transparent
!
vlan 15
```

```
name Lane-Safety-Systems
vlan 25
name Staff-POS
vlan 45
name IT-Mgmt
vlan 925
name Guest
vlan 999
name Native-Unused
exit
!
spanning-tree mode rapid-pvst
spanning-tree vlan 15,25,45,925,999 priority 12288
!
interface fastEthernet0/1
description Trunk to SW-HarborDistrict Fa0/3
switchport mode trunk
switchport trunk native vlan 999
switchport trunk allowed vlan 15,25,45,925,999
switchport nonegotiate
exit
interface fastEthernet0/2
description Trunk to SW-Downtown Fa0/3
switchport mode trunk
switchport trunk native vlan 999
switchport trunk allowed vlan 15,25,45,925,999
switchport nonegotiate
exit
!
interface range fastEthernet0/5 - 10
description Lane-Safety-Systems (sensors, scoring terminals)
switchport mode access
switchport access vlan 15
spanning-tree portfast
exit
interface range fastEthernet0/11 - 14
description Staff-POS
switchport mode access
switchport access vlan 25
spanning-tree portfast
exit
interface range fastEthernet0/15 - 16
description IT-Mgmt (AP management)
switchport mode access
switchport access vlan 45
exit
interface range fastEthernet0/17 - 24
description Guest lobby ports (port-security)
switchport mode access
switchport access vlan 925
switchport port-security
switchport port-security maximum 2
switchport port-security violation restrict
switchport port-security mac-address sticky
spanning-tree portfast
spanning-tree bpduguard enable
exit
interface range fastEthernet0/3 - 4 , gigabitEthernet0/1 - 2
description UNUSED
```

```
switchport mode access
switchport access vlan 999
shutdown
exit
!
interface vlan 45
ip address 10.122.45.12 255.255.255.0
no shutdown
exit
ip default-gateway 10.122.45.1
end
write memory
```

A.4 HD-Router

```
enable
configure terminal
hostname HD-Router
no ip domain-lookup
!
ip access-list extended GUEST_IN
remark Guest may never reach any internal VLAN
deny ip 10.122.200.0 0.0.0.255 10.122.0.0 0.0.127.255
permit ip any any
exit
ip access-list extended LANE_OUT
remark Only Staff-POS and IT-Mgmt may reach Lane-Safety-Systems
permit ip 10.122.25.0 0.0.0.255 10.122.15.0 0.0.0.255
permit ip 10.122.45.0 0.0.0.255 10.122.15.0 0.0.0.255
deny ip any any
exit
!
interface gigabitEthernet0/0
description Trunk to SW-HarborDistrict Fa0/4
no ip address
no shutdown
exit
interface gigabitEthernet0/0.15
description Lane-Safety-Systems gateway
encapsulation dot1Q 15
ip address 10.122.15.1 255.255.255.0
ip access-group LANE_OUT out
exit
interface gigabitEthernet0/0.25
description Staff-POS gateway
encapsulation dot1Q 25
ip address 10.122.25.1 255.255.255.0
exit
interface gigabitEthernet0/0.35
description Admin gateway
encapsulation dot1Q 35
ip address 10.122.35.1 255.255.255.0
exit
interface gigabitEthernet0/0.45
description IT-Mgmt gateway
encapsulation dot1Q 45
ip address 10.122.45.1 255.255.255.0
exit
interface gigabitEthernet0/0.925
```

```
description Guest gateway
encapsulation dot1Q 925
ip address 10.122.200.1 255.255.255.0
ip access-group GUEST_IN in
exit
interface gigabitEthernet0/0.999
description Native VLAN (unused, no IP)
encapsulation dot1Q 999 native
exit
end
write memory
```

END OF REPORT